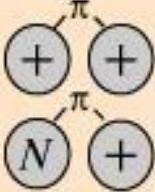
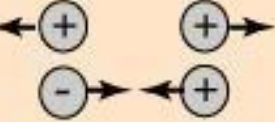
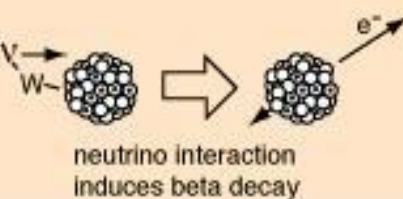
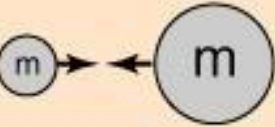


# Introduction

Electromagnetic Fields and Waves

# Electromagnetic interaction is ubiquitous

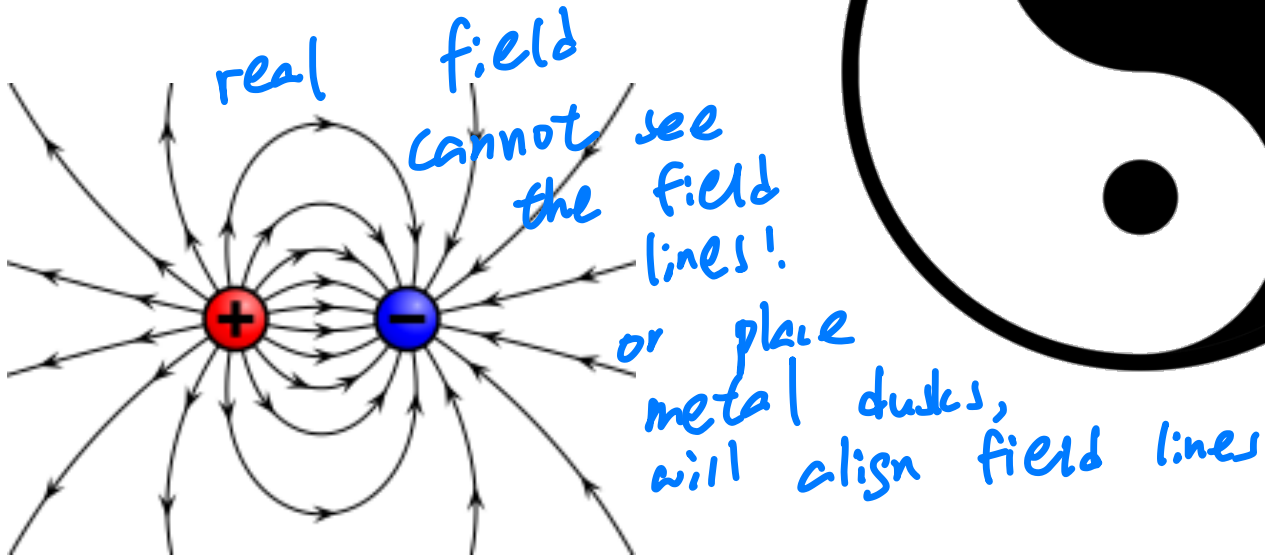
| Fundamental Forces      |                                                                                                                                    |                                 |                                                                 |                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <i>Strong</i>           |  <p>Force which holds nucleus together</p>        | Strength<br><b>1</b>            | Range (m)<br>$10^{-15}$<br>(diameter of a medium sized nucleus) | Particle<br>gluons,<br>$\pi$ (nucleons)                                                        |
| <i>Electro-magnetic</i> |                                                  | Strength<br>$\frac{1}{137}$     | Range (m)<br>Infinite                                           | Particle<br>photon<br>mass = 0<br>spin = 1                                                     |
| <i>Weak</i>             |  <p>neutrino interaction induces beta decay</p> | Strength<br>$10^{-6}$           | Range (m)<br>$10^{-18}$<br>(0.1% of the diameter of a proton)   | Particle<br>Intermediate vector bosons<br>$W^+$ , $W^-$ , $Z_0$ ,<br>mass > 80 GeV<br>spin = 1 |
| <i>Gravity</i>          |                                                | Strength<br>$6 \times 10^{-39}$ | Range (m)<br>Infinite                                           | Particle<br>graviton ?<br>mass = 0<br>spin = 2                                                 |

*We can smell!*

# Electricity and magnetism



Electricity



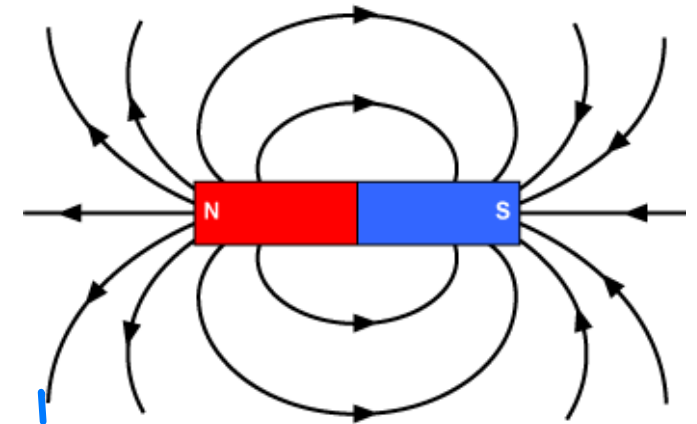
Electric field:  $\vec{E}$

Light is EM field



Magnetism

dipole!



Magnetic field:  $\vec{B}$

# Maxwell's Equations

*will learn in the entire course!*

| Name                                                     | Integral equations                                                                                                                                                                             | Differential equations                                                                                                               |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Gauss's law                                              | $\oiint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon_0} \iiint_{\Omega} \rho dV$                                                                                          | <i>divergence</i> $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ <i>← electron density at this particular time &amp; space!</i> |
| Gauss's law for magnetism                                | $\oiint_{\partial\Omega} \mathbf{B} \cdot d\mathbf{S} = 0$                                                                                                                                     | $\nabla \cdot \mathbf{B} = 0$                                                                                                        |
| Maxwell–Faraday equation<br>(Faraday's law of induction) | $\oint_{\partial\Sigma} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}$                                                                              | $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$                                                                 |
| Ampère's circuital law (with<br>Maxwell's addition)      | $\oint_{\partial\Sigma} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left( \iint_{\Sigma} \mathbf{J} \cdot d\mathbf{S} + \epsilon_0 \frac{d}{dt} \iint_{\Sigma} \mathbf{E} \cdot d\mathbf{S} \right)$ | $\nabla \times \mathbf{B} = \mu_0 \left( \mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$                     |

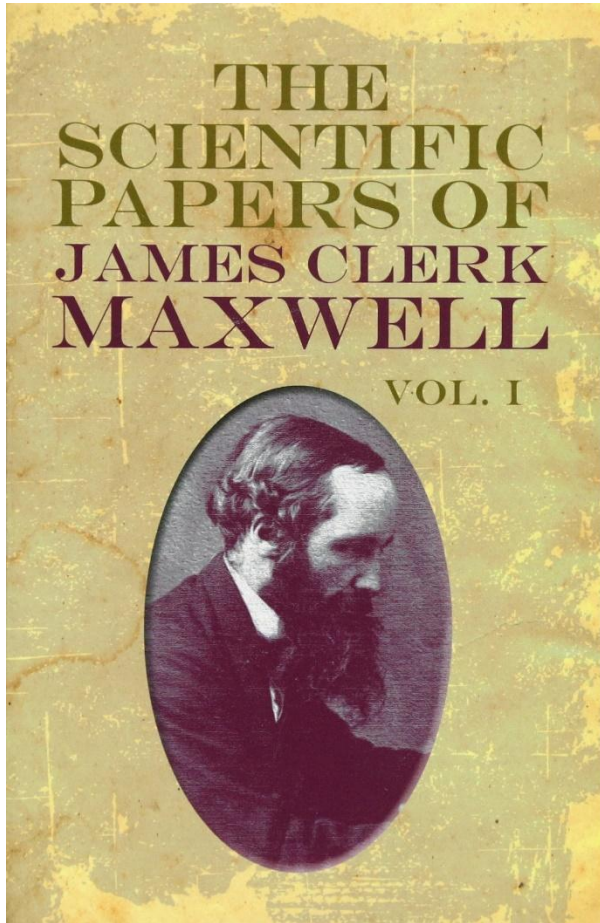
*→ difficult to solve!*

- A **set** of **coupled** linear **partial** differential equations of **vectors**!
- They dictate everything about classical electromagnetism.

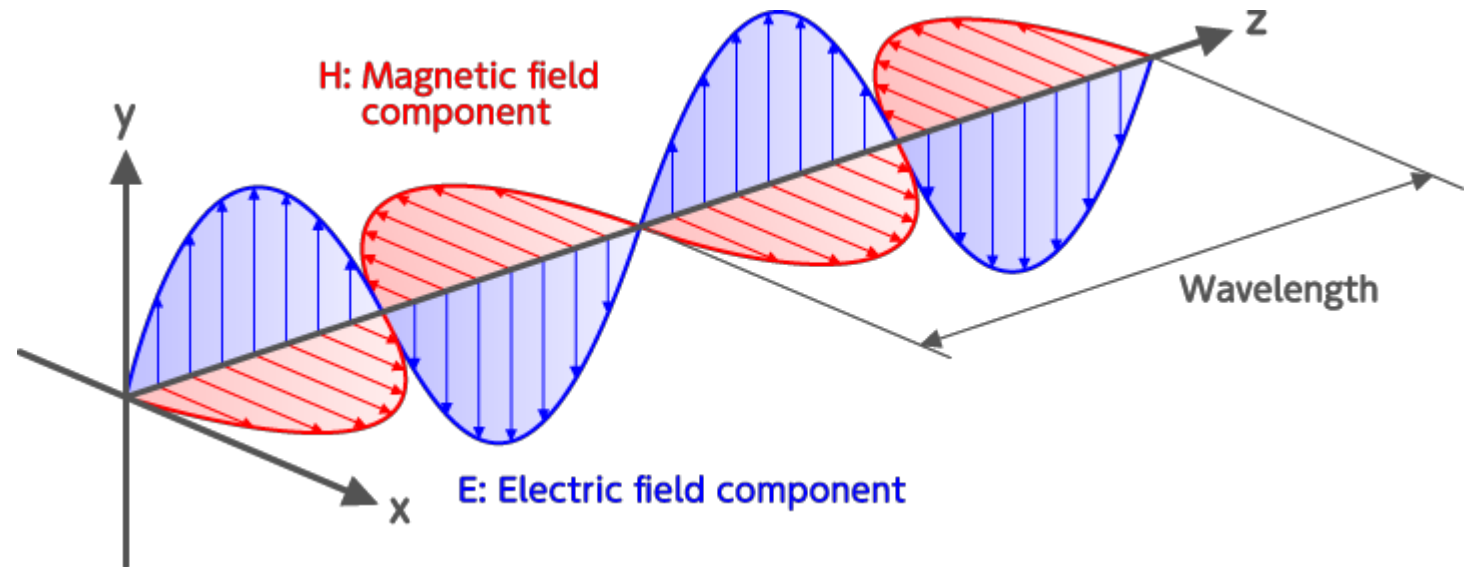
*↓  
8 equation  
actually total 20! <sup>4</sup>*

# A glorious moment in science history

will spend one <sup>entire</sup> lecture for that!



Published 1856 ~ 1862



*light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena*

*--James Clerk Maxwell*

# It is a “classical” theory

- “Classical” means: it does not incorporate quantum principles (e.g., photon quantization)
- “Classical” also means: it lays the foundation for modern physics. **Maxwell theory introduced:**

- Field as fundamental entity
- Local differential equations
- Finite propagation speed
- Lorentz invariance
- Gauge redundancy and gauge invariance

→ in deep'

empty  
charge / field  
with magnet

They are structural pillars of physics.

↳ important concepts



# It is a “classical” theory (contd.)

- **Maxwell theory directly led to:**

- Special Relativity
- The concept of relativistic field theory

- **Maxwell theory later generalized into:**

- Quantum Electrodynamics
- Gauge theory

- <https://www.bilibili.com/video/av17923311/>

- Chen Ning Yang, “The conceptual origins of Maxwell’s equations and gauge theory”, Physics Today **67**, 11, 45 (2014)

- Gauge symmetry originates in electromagnetism, not quantum mechanics.
- Maxwell theory introduced local gauge invariance — the organizing principle of modern particle physics.

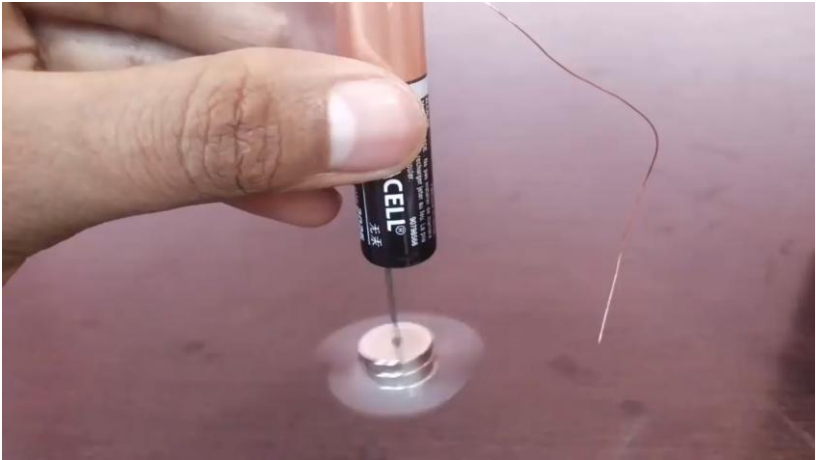
# Join the tour of Maxwell's wonder world

Maxwell was a religious person.  
I wonder whether he had in  
his prayers asked for God's  
forgiveness for revealing one  
of His greatest secrets.

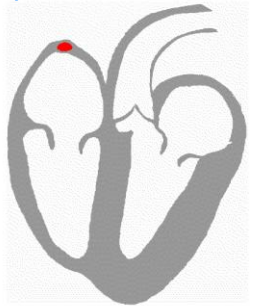
-- Chen Ning Yang



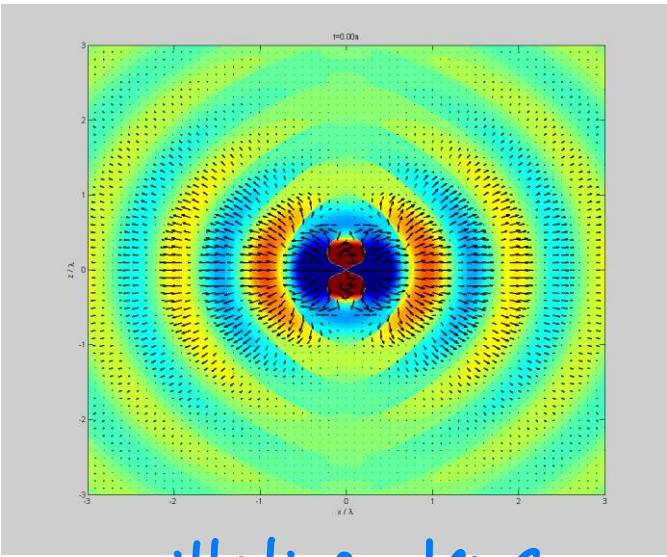
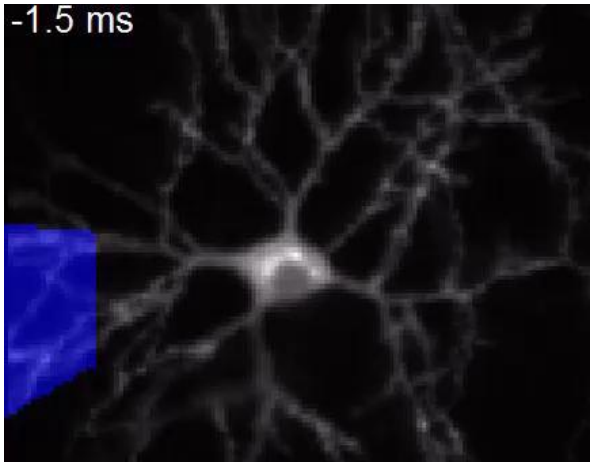
# Interesting EM phenomena



heartbeats



neurons



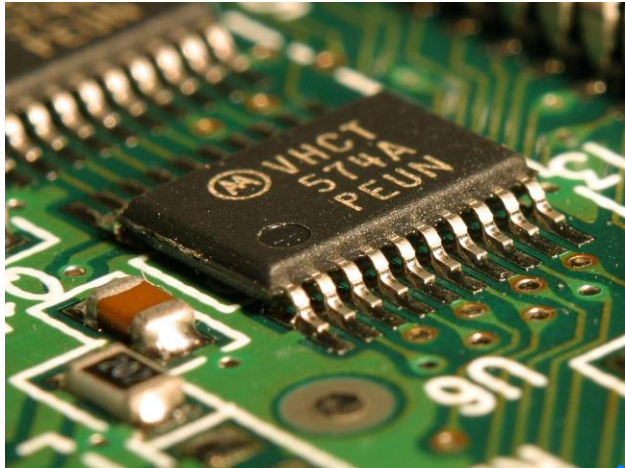
oscillating charge



float



# Electromagnetism is useful



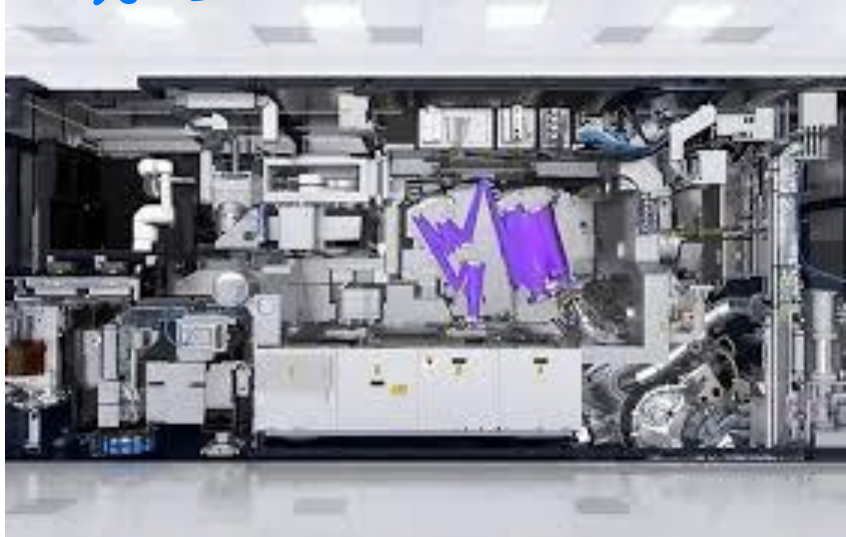
电子弹射



no dem imaging



光刻



Tencent (VooV) Meeting ID: 529-9925-0249

## Instructor

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Contact: cheng\_ma@tsinghua.edu.cn (e-mail), 15652659600 (cell, country code: +86).

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## Office Hours

Thursdays, 2:00 to 4:00 pm (appointments needed).

## • Textbooks

1. Main textbook: *Introduction to Electrodynamics*, David J. Griffiths, Cambridge University Press, Fourth Edition (ISBN: 978-1108420419).

- (Electronic version downloadable at <http://reserves.lib.tsinghua.edu.cn/Home>).

2. *The Feynman Lectures on Physics, Volume II*

- (Electronic version downloadable at [https://www.feynmanlectures.caltech.edu/II\\_toc.html](https://www.feynmanlectures.caltech.edu/II_toc.html))

3. 《电动力学》郭硕鸿 著，高等教育出版社，第三版(ISBN:978-7040239249).

- **Grading**

- Homework 20%; midterm exam 40%; final exam 40%.

- **A (A, A-): 20%; B (B+, B, B-): 50%; else: 30%.**

| Week | Date   | Topics                                                                                                         |
|------|--------|----------------------------------------------------------------------------------------------------------------|
| 1    | Feb 27 | Basics of vector analysis.                                                                                     |
| 2    | Mar 6  | Electrostatics, Coulomb's law, Gauss's law, conductors.                                                        |
| 3    | Mar 13 | Electric potential, dielectric and polarization.                                                               |
| 4    | Mar 20 | Electric boundary conditions, uniqueness theorem, method of images, separation of variables.                   |
| 5    | Mar 27 | Magnetostatic fields, steady currents.                                                                         |
| 6    | Apr 3  | Properties of magnetostatic fields, vector potential, magnetic fields in matter, magnetic boundary conditions. |
| 7    | Apr 10 | Force and energy of electric and magnetic dipoles, A-B effect, electromagnetic induction, Faraday's law.       |
| 8    | Apr 17 | Displacement current, Maxwell's equations.                                                                     |
| 9    | Apr 24 | Mid-term exam (open-book, 2 hours).                                                                            |
| 10   | May 1  | No class                                                                                                       |
| 11   | May 8  | Poynting's theorem, EM waves, plane waves.                                                                     |
| 12   | May 15 | General solutions to the wave equations, dipole radiation, antenna.                                            |
| 13   | May 22 | Linear light-matter interaction, the origin of refractive index, absorption, dispersion.                       |
| 14   | May 29 | Reflection and transmission of plane waves at an interface.                                                    |
| 15   | Jun 5  | Waveguides.                                                                                                    |
| 16   | Jun 12 | Course review or no class, depending on the final exam date                                                    |
| 17   | TBD    | Final exam (open-book, 2 hours).                                                                               |

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

**Fields**



**Waves**



# Hints about how to study this course

- Be prepared:
  - Not intuitive (compared to classical mechanics)
  - Very abstract (field, potentials, **vector potential**, waves...)
- Keys: **imagination** and **mathematics**

## Physics



VS.

## Math



- Study tip: Preview the material and bring your questions to class.

Brief review of basic math:

1.  $a, b, a+b, ab=ba, a(b+c) = ab + ac, (ab)c = a(bc)...$

2.  $f(x), f'(x), df(x) = f'(x)dx, (fg)' = f'g + fg'...$

3.  $f''(x) = (f'(x))'$

4.  $\int_a^b f'(x)dx = f(b) - f(a)$

$$\vec{A} = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x \ A_y \ A_z) \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \\ &= A_x B_x + A_y B_y + A_z B_z \\ &= \text{scalar} \end{aligned}$$

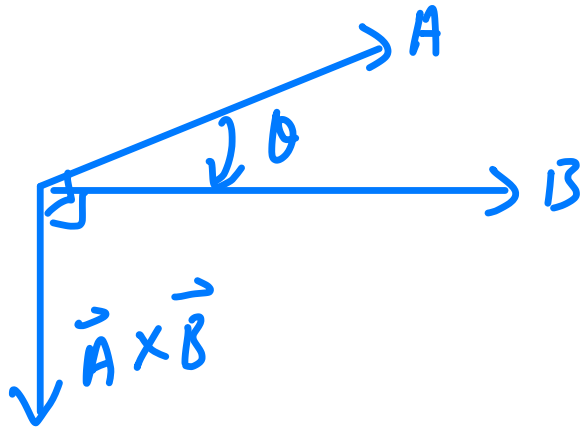
$$= AB \cos \theta \quad , \text{ if } \theta = 90^\circ, \text{ independent}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$$



# Triple product

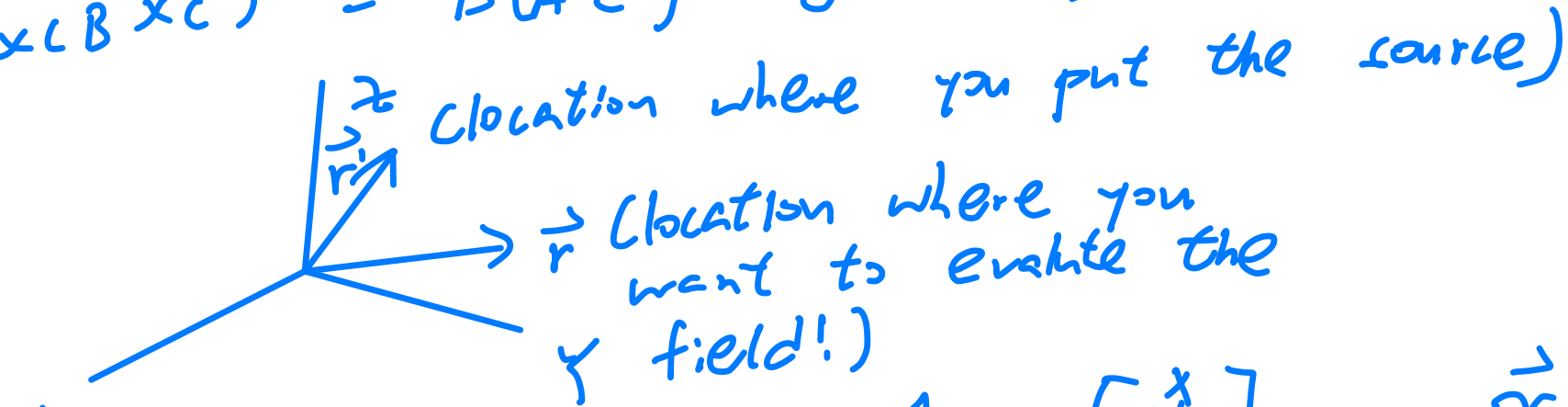
$$\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

Rotation!

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

BAC CAB Rule



$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\vec{r}' = x'\hat{x} + y'\hat{y} + z'\hat{z} = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}$$

$$\vec{r} = \begin{bmatrix} x - x' \\ y - y' \\ z - z' \end{bmatrix}$$

$$\hat{r} = \dots$$

$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z} \quad (\text{divide to components})$$

Differential Cal.

$$f(x) \rightarrow f'(x)$$

$$\nabla \text{ del operator } \triangleq \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}$$

$$\frac{d}{dx}$$

Gradient

$$\nabla f(x, y, z) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{bmatrix} \quad \text{vector function}$$

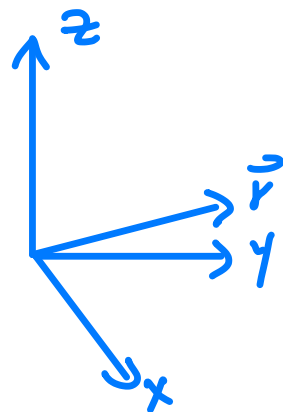
$$df(x, y, z) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \cdot \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$

$$= \nabla f \cdot d\vec{l}$$

$\nabla f \cdot d\vec{l}$   
along direction, what change is  
the maximum

$$d\vec{l} = \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$



$$\begin{aligned}
 \nabla h &= (\partial_x, \partial_y, \partial_z) \overset{\text{mark!}}{(\sqrt{x^2+y^2+z^2})} \\
 &= \left\langle \frac{2x}{2r}, \frac{2y}{2r}, \frac{2z}{2r} \right\rangle \\
 &= \left\langle \frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right\rangle \\
 &= \frac{1}{r} \langle x, y, z \rangle \\
 &= \hat{r}
 \end{aligned}$$

Divergence

$$\nabla \cdot \vec{f} = (\partial_x, \partial_y, \partial_z) \cdot \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix}$$

$$= \partial_x f_x + \partial_y f_y + \partial_z f_z$$

No term like  $\partial_x f_y$

$$\nabla \cdot \frac{\hat{r}}{r^2} =$$

most fundamental function  
of entire universe!

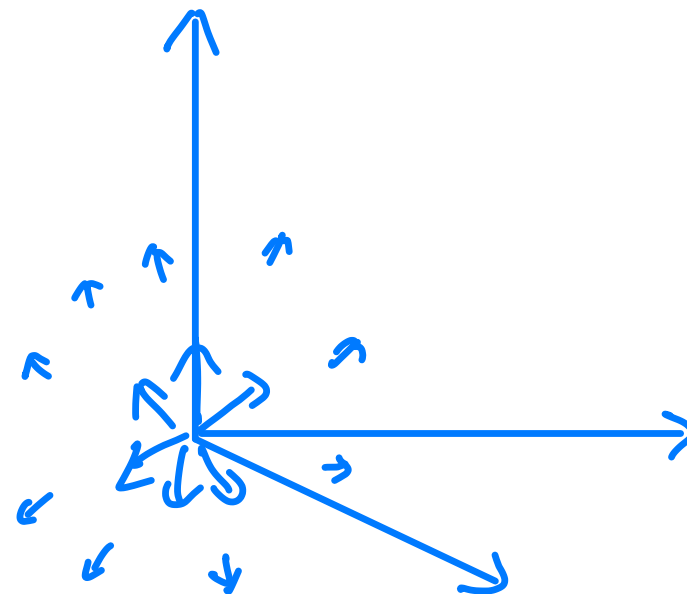
like a fountain

How similar the  
field like this  
pattern, so taking  
dot product.





$$\begin{aligned}
 \nabla \frac{1}{r^2} &= \nabla \cdot \frac{\vec{r}}{r^3} \\
 &= (\partial_x, \partial_y, \partial_z) \cdot \frac{\langle x, y, z \rangle}{r^3} \\
 &= \frac{1}{r^3} - \frac{3}{2} \frac{2x^2}{r^3} + \dots
 \end{aligned}$$

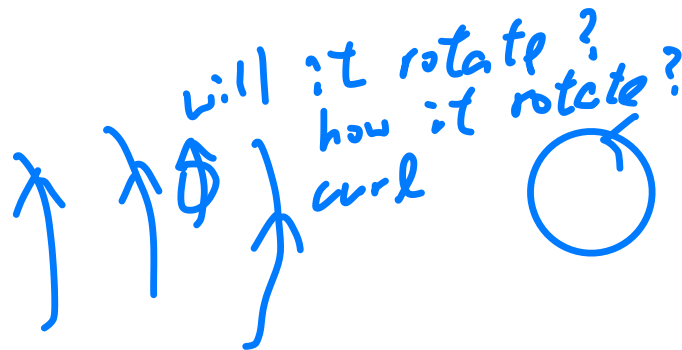


Curl

$$\nabla \times \vec{f} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_x & f_y & f_z \end{vmatrix}$$

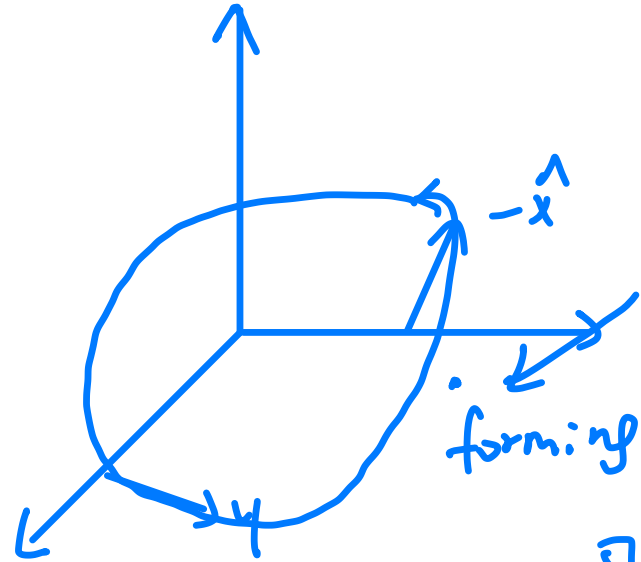
$$= \hat{x} (\partial_y f_z - \partial_z f_y) + \hat{y} (\partial_z f_x - \partial_x f_z) + \hat{z} (\partial_x f_y - \partial_y f_x)$$

fixed location, how it rotating around?



$\nabla \times \vec{f} > 0$   
ball will not rotate!

$$\vec{f} = x\hat{y} - y\hat{x}$$



curl at this location still 2!

forming a loop

$$\nabla \times \vec{f} = \dots (\leq 1)$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & 0 \end{vmatrix} = 2\hat{z}$$

Right hand rule, counterclockwise along x-y plane!

Now move on to 2nd derivative  
 $f'' = ?$

↑  
use del operator

$$\nabla \cdot (\nabla f) = (\partial_x \ \partial_y \ \partial_z) \begin{bmatrix} \partial_x f \\ \partial_y f \\ \partial_z f \end{bmatrix}$$

$$= \partial_x \partial_x f + \partial_y \partial_y f + \partial_z \partial_z f$$

$$= (\partial_x^2 + \partial_y^2 + \partial_z^2) f$$

$$= \nabla^2 f$$

↑  
scalar operator!

$$\nabla^2 \vec{f} = \begin{bmatrix} \nabla^2 f_x \\ \nabla^2 f_y \\ \nabla^2 f_z \end{bmatrix}$$

$$\nabla \times (\nabla \times \vec{f})$$

$$= \nabla (\nabla \cdot \vec{f}) - \nabla^2 \vec{f} \quad (\text{remember this result})$$

→ derive wave equation

BAC - CAB rule!

Once a vector function can  
be represented by gradient of  
scalar function, curl will be vanished

$$\nabla \times (\nabla f)$$

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ f_x & f_y & f_z \end{vmatrix}$$

$$= \hat{x} (\partial_y \partial_z f - \partial_z \partial_y f) + \dots$$

$$= \vec{0}$$

★ very important!

Because  $\nabla f$  maybe very complex!

nontrivial

$$\nabla \cdot (\nabla \times \vec{f})$$

$$= \begin{pmatrix} \partial_x & \partial_y & \partial_z \end{pmatrix} \begin{bmatrix} \partial_y f_z - \partial_x f_y \\ \partial_z f_x - \partial_x f_z \\ \partial_x f_y - \partial_y f_x \end{bmatrix}$$

$$= \partial_x \partial_y f_z - \partial_x \partial_x f_y + \dots$$

$$= 0$$

divergence and curl is 0

$\nabla \vec{f}$ ,  $\nabla$  is perpendicular!  $\Rightarrow$  no magnetic monopole



# Integral Calculus

$$\int_a^b f'(x) dx = f(b) - f(a)$$

$f'(x)$  may be complex function

- not a trivial result

$$\int_{\vec{a}}^{\vec{b}} \nabla f(x, y, z) \cdot d\vec{l} \quad \begin{matrix} \rightarrow \text{field, conservative} \\ \text{field} \end{matrix}$$

$$= \int_{\vec{a}}^{\vec{b}} df = f(\vec{b}) - f(\vec{a}) \quad \text{much more}$$

1. Independent on path!

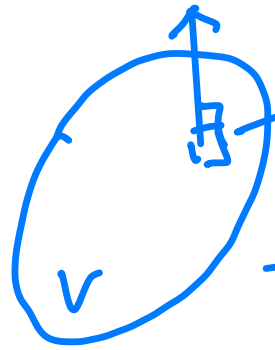
2. curl of such function  $= 0$

Same final result



$$\iiint_V \nabla \cdot \vec{f} dV \quad \text{volume integration}$$

$\downarrow$



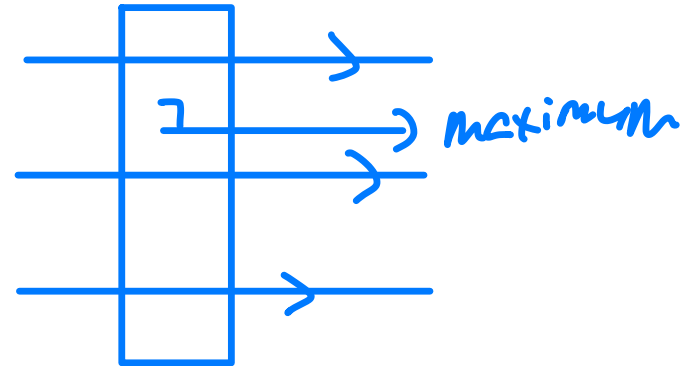
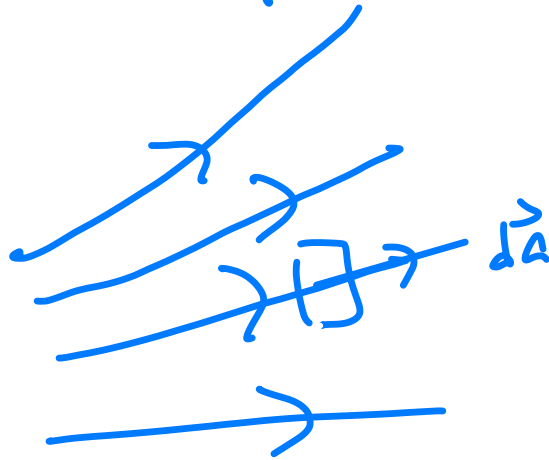
$d\vec{a}$   
direction  $\perp$  local area

$- S$ , closed surface

$$= \oint_S \vec{f} \cdot d\vec{a} = \text{flux}$$

closed surface

But not easy to prove



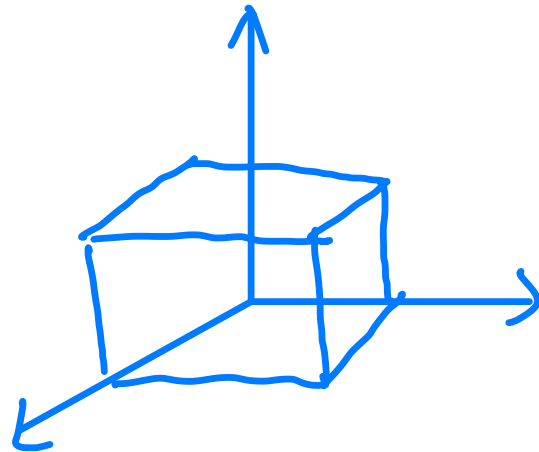
$\int$ , Total number of field lines that penetrates the surface

# Divergence theorem

$$\int_{z_1}^{z_2} \int_{y_1}^{y_2} \int_{x_1}^{x_2} \nabla \cdot f(x, y, z) dx dy dz$$

$$\downarrow$$

$$\partial_x f_x + \partial_y f_y + \partial_z f_z$$



$$= \int_{z_1}^{z_2} \int_{y_1}^{y_2} f(x_2, y, z) dy dz -$$

$$\int_{z_1}^{z_2} \int_{y_1}^{y_2} f(x_1, y, z) dy dz + \dots$$

(4 terms left)

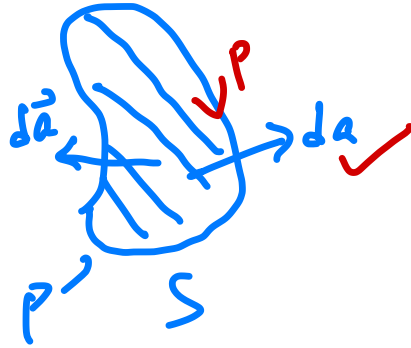
$$= \int_{S_1} \vec{f} \cdot d\vec{a} + \int_{S_2} \vec{f} \cdot d\vec{a} + \dots$$

$\uparrow$   
 $dy dz$ , along  $\hat{x}$

$\uparrow$   
 4 surfaces

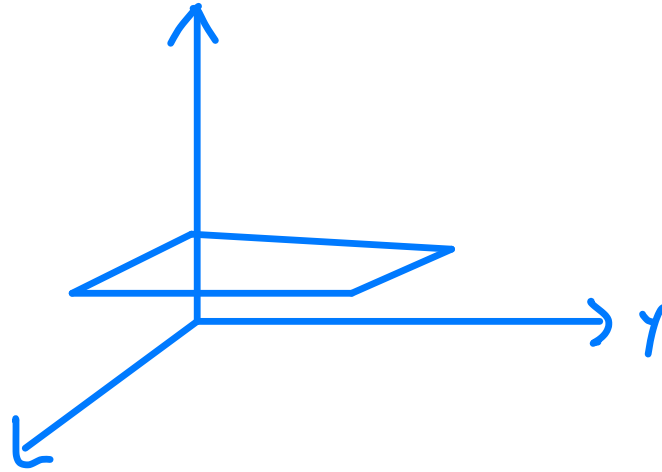
Stoke's theorem

$$\iint_S (\nabla \times \vec{f}) \cdot d\vec{a} = \oint_P \vec{f} \cdot d\vec{\ell}$$



open surface

$P$  is defined by  
right hand rule



$$\nabla \cdot \vec{E} = \iiint_V \nabla \cdot \vec{E} \, dV = \frac{\iiint_V \rho(x, y, z) \, dV}{\epsilon_0} = \frac{Q_{enc}}{\epsilon_0}$$

Divergence theorem

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

Total flux

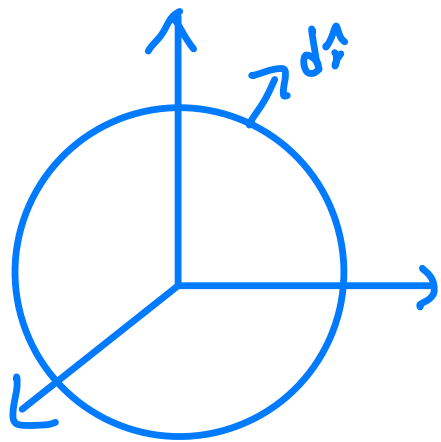
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\iint_S \nabla \times \vec{E} \cdot d\vec{a} = -\frac{d}{dt} \underbrace{\iint_S \vec{B} \cdot d\vec{a}}_{\text{flux}} = -\frac{d}{dt} \Phi_B = \oint \vec{E} \cdot d\vec{\ell}$$

电动势

- only study local behaviour in flux

$$\iiint_V \nabla \cdot \frac{\hat{r}}{r^2} dV = \oiint_S \frac{\hat{r} \cdot \hat{n}}{R^2} R^2 \sin \theta d\theta d\phi$$



$$= \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi$$

$$= 2\pi \cdot 2 = 4\pi$$

$\neq 0$

origin is not 0!

$$\left( \nabla \frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(\vec{r})$$

origin

$$\delta^3(\quad) = \delta^3(\vec{r} - \vec{r}') \text{ shift the source!}$$

$$\delta(x) = \begin{cases} \text{impulse} & x=0 \\ 0 & x \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$